

Exploratory Data Analysis (EDA) on Netflix Dataset

1. Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on the Netflix Titles dataset sourced from Kaggle. The dataset contains information about Netflix's content library, including attributes such as title, type (Movie or TV Show), country of origin, genre, rating, duration, date added, and release year.

The main objective of this project is to analyze Netflix's content distribution and uncover meaningful patterns and trends using data cleaning techniques and visualizations. The analysis helps in understanding how Netflix content has evolved over time and what types of content are most dominant on the platform.

2. Data Loading and Initial Exploration

The dataset was loaded using Pandas and initially inspected using basic functions such as `head()`, `shape`, `columns`, and `info()` to understand its structure.

During exploration:

- The dataset contained multiple columns including both categorical and numerical features.
- Unnecessary unnamed columns were removed to clean the dataset.
- The structure of the dataset was analyzed to understand missing values, data types, and overall size.

This step helped in gaining a clear understanding of the dataset before performing deeper analysis.

3. Data Cleaning and Preprocessing

Data cleaning was an important step to ensure accuracy and consistency in analysis.

The following preprocessing steps were performed:

- Removed unwanted or unnamed columns.
- Handled missing values by replacing them with "Unknown" to avoid data loss.

- Converted the `date_added` column into datetime format for time-based analysis.
- Created a new feature `year_added` from the `date_added` column to analyze yearly trends.
- Standardized text-based fields for better grouping and visualization.

These steps ensured that the dataset was clean, structured, and ready for visualization and analysis.

4. Exploratory Data Analysis and Visualizations

4.1 Movies vs TV Shows Distribution

A pie chart (donut chart) was used to visualize the distribution of Movies and TV Shows on Netflix.

Insight:

- Movies dominate the Netflix platform compared to TV Shows.
 - This indicates that Netflix has a larger library of movies than series content.
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4.2 Content Added Over the Years

A line chart was used to analyze the trend of content added over time.

Insight:

- There is a significant increase in content addition after 2015.
 - Netflix experienced rapid expansion in recent years, increasing its content library aggressively.
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4.3 Top Countries Contributing Content

A world map (choropleth map using Plotly) was used to visualize content distribution by country.

Insight:

- The United States contributes the highest number of Netflix titles.
 - Other countries contribute significantly less in comparison, showing a strong dominance of US-produced content.
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4.4 Content Rating Distribution

A count plot was used to analyze rating distribution.

Insight:

- TV-MA is one of the most common ratings.
 - This indicates that a large portion of Netflix content is intended for mature audiences.
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4.5 Top Genres on Netflix

A bar chart was used to analyze the most common genres.

Insight:

- Drama is the most dominant genre on Netflix.
 - Other popular genres include Comedy and International content.
 - Netflix offers a wide variety of genres but Drama remains the most frequent.
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4.6 Movie Duration Analysis

A histogram was used to analyze the duration of movies.

Insight:

- Most movies fall within a standard duration range.
 - The average movie duration is around a typical feature-length format.
 - This shows consistency in movie length across the platform.
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5. Hypothesis Testing

Before analysis, four hypotheses were defined and tested using visualizations and statistics.

Hypothesis 1: Movies are more than TV Shows

- **Method Used:** Value counts + Pie chart (donut chart)
 - **Statistical Check:** Count of Movies vs TV Shows using `df['type'].value_counts()`
 - **Visualization:** Pie chart showing proportion of Movies and TV Shows
 - **Result:** Movies are significantly higher than TV Shows
 - **Conclusion:** Hypothesis is **validated**
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Hypothesis 2: Most content was added after 2015

- **Method Used:** Time-series analysis using `year_added`
 - **Statistical Check:** Value counts of content added per year
 - **Visualization:** Line chart showing yearly content addition trend
 - **Result:** Sharp increase in content after 2015
 - **Conclusion:** Hypothesis is **validated**
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Hypothesis 3: Drama is the most frequent genre

- **Method Used:** Genre extraction using `listed_in` column
 - **Statistical Check:** Frequency count of all genres using value counts
 - **Visualization:** Bar chart of top 10 genres
 - **Result:** Drama appears most frequently among all genres
 - **Conclusion:** Hypothesis is **validated**
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Hypothesis 4: Movie durations follow a consistent pattern

- **Method Used:** Numerical transformation of duration column
 - **Statistical Check:** Average movie duration calculation using `.mean()`
 - **Visualization:** Histogram with KDE curve
 - **Result:** Most movies fall within a standard duration range
 - **Conclusion:** Hypothesis is **validated**
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6. Key Insights

- Netflix has more Movies than TV Shows.
 - The United States is the leading contributor of Netflix content.
 - Content production increased significantly after 2015.
 - Drama is the most dominant genre on Netflix.
 - Most content is targeted toward mature audiences (TV-MA rating).
 - Movie durations follow a consistent pattern.
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7. Detection of Potential Data Issues

During Exploratory Data Analysis, several data quality issues were identified and handled to improve accuracy of analysis.

1. Missing Values

- Several columns such as country, director, rating, and cast contained missing values.
 - Action Taken: Missing values were replaced with "Unknown" to avoid analysis errors.
 - Impact: Ensured completeness of dataset but introduced generic placeholder values.
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2. Unnamed / Irrelevant Columns

- The dataset contained unwanted unnamed columns.
 - Action Taken: These columns were removed using column filtering.
 - Impact: Improved dataset clarity and reduced noise.
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3. Inconsistent Date Format

- The date_added column was not in proper datetime format.
 - Action Taken: Converted using pd.to_datetime().
 - Impact: Enabled time-series analysis and yearly trend extraction.
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4. Multiple Countries in Single Cell

- The country column contained multiple countries separated by commas.
 - Action Taken: Split and stacked values for proper aggregation.
 - Impact: Improved accuracy of country-level analysis but increased data complexity.
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5. Genre Column Complexity

- The listed_in column contained multiple genres per entry.
 - Action Taken: Split genres and calculated frequency distribution.
 - Impact: Allowed multi-label genre analysis but required additional preprocessing.
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6. Duration Format Issue

- Movie duration was stored as text (e.g., "90 min").
- Action Taken: Removed text and converted into numeric format.

- Impact: Enabled statistical analysis of movie durations.
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7. Data Skewness (Imbalance)

- Dataset showed imbalance in:
 - Movies vs TV Shows
 - Country distribution (US dominance)
- Impact: May bias conclusions toward dominant categories.

8. Conclusion:

This Exploratory Data Analysis project successfully analyzed Netflix's content dataset and uncovered meaningful insights using Python-based data analysis and visualization techniques.

The project demonstrates how raw data can be transformed into useful information through cleaning, preprocessing, and visualization. The analysis revealed important trends such as the dominance of movies, rapid content growth after 2015, and genre preferences among users.

By combining hypothesis testing with visual analytics, this project provides a strong understanding of Netflix's content strategy and user preferences. It also highlights the importance of EDA in extracting actionable insights from real-world datasets.

Overall, this project strengthens skills in Python, Pandas, Matplotlib, Seaborn, and data storytelling, making it a valuable addition to a data analytics portfolio.